

Counterfactual Inflation Analysis: What If Ukraine Had Been Part of the Euro Area?

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Abstract

This paper constructs a counterfactual inflation trajectory for Ukraine under the hypothesis that it had been a member of the Euro Area between 2006 and 2025. We first document the time-varying nature of Ukraine’s monetary sovereignty, showing that the National Bank of Ukraine (NBU) operated a *de facto* U.S. dollar peg over much of the period and only exercised genuine monetary autonomy during the 2008–09 and 2014–15 devaluation episodes and the post-2016 inflation-targeting regime. Building on this chronology, we estimate a bivariate Bayoumi–Eichengreen (1993) Structural VAR identified by the long-run restriction of Blanchard and Quah (1989) on output and inflation, separately for Ukraine and the Euro Area. The counterfactual inflation series is constructed by *shock replacement*: Ukraine retains its own idiosyncratic supply shocks while inheriting the Euro Area’s monetary anchor. The result is then weighted by a time-varying treatment intensity θ_t that reflects the actual degree of NBU monetary autonomy. We find that Euro Area membership would have substantially reduced Ukrainian inflation: the average gap between actual and counterfactual inflation reaches +27 percentage points during the 2014–15 currency collapse and +12 percentage points during the 2016–19 inflation-targeting period, while remaining negligible during the *de facto* peg years. The cost of monetary sovereignty for Ukraine is therefore concentrated in episodes when this sovereignty was actively used, primarily through devaluations of the hryvnia.

Keywords: Counterfactual analysis, Structural VAR, Blanchard–Quah identification, Optimal Currency Areas, Monetary sovereignty, Ukraine, Euro Area.

1 Introduction

On 23 June 2022, the European Council granted Ukraine official candidate status for accession to the European Union. Although EU membership and Euro Area membership are distinct steps, separated by years of institutional convergence and the fulfilment of the Maastricht criteria, the question of whether Ukraine could eventually adopt the euro has moved from purely academic to a question that the European Central Bank (ECB), the Eurosystem and the National Bank of Ukraine (NBU) will increasingly need to assess.

This paper addresses the foundational counterfactual question: *What would Ukraine’s inflation trajectory have looked like had it been a member of the Euro Area?*

Ukraine was never in the Euro Area. It maintained its own currency (the hryvnia, UAH), its own central bank (the NBU) and its own monetary policy throughout the sample. The exercise is therefore a counterfactual construction in the sense of [Abadie et al. \(2010\)](#) and the broader potential-outcomes framework of [Angrist and Pischke \(2009\)](#), but applied here through a structural macroeconometric lens.

The economic stakes of this question follow directly from the theory of Optimal Currency Areas ([Mundell, 1961](#); [McKinnon, 1963](#); [Kenen, 1969](#)): joining a monetary union trades off the loss of independent monetary and exchange-rate policy against the gains from imported credibility, reduced transaction costs and price transparency. For Ukraine, this trade-off is particularly stark: the country experienced three major hryvnia devaluations ($\sim -50\%$ in 2008–09, -200% in 2014–15, -25% in 2022–23) which acted as a critical adjustment mechanism, but suffered persistently high inflation (mean of 12.8% year-on-year over the sample, peaking at 60.9% in April 2015) consistent with weak monetary credibility ([Barro and Gordon, 1983](#); [Giavazzi and Pagano, 1988](#)).

The paper makes three contributions. First, in Section 2 we build a documented chronology of Ukraine’s exchange rate regimes from 2000 to 2025 and operationalise it as a time-varying monetary sovereignty index $\theta_t \in [0, 1]$. This index is essential to the identification strategy: as [Calvo and Reinhart \(2002\)](#) emphasise, *de jure* monetary sovereignty can diverge sharply from *de facto* sovereignty, and any meaningful counterfactual must reflect this asymmetry. Second, in Sections 4–5 we estimate a bivariate Structural VAR à la [Bayoumi and Eichengreen \(1993\)](#) with the long-run identification scheme of [Blanchard and Quah \(1989\)](#), separately for Ukraine and the Euro Area aggregate. We extract structural supply and demand shocks and verify that they are economically interpretable: the supply-shock series exhibits the expected spikes around the 2008 crisis, the 2014–15 conflict and the 2022 invasion, while demand shocks capture monetary regime transitions. Third, in Section 5 we construct the counterfactual via a shock-replacement scheme grounded in the OCA logic and weight it by θ_t to ensure consistency with Part A. Section 6 presents and interprets the empirical results.

The remainder of the paper is organised as follows. Section 2 presents the analysis of Ukraine’s monetary regime. Section 3 describes the data. Section 4 details the model specification and identification strategy. Section 5 explains the construction of the counterfactual. Section 6 presents and interprets the results. Section 7 provides robustness and sanity checks. Section 8 concludes.

2 Part A — Ukraine’s Monetary Regime: A Preliminary Analysis

A naive framing of the counterfactual would assume that Ukraine operated under a freely floating exchange rate with full monetary policy independence throughout the sample period. This view is incorrect. Following [Calvo and Reinhart \(2002\)](#), who document the widespread phenomenon of “fear of floating” among emerging-market central banks, we establish that the NBU’s *de facto* regime departed substantially from its *de jure* classification over much of the period.

2.1 Chronology of Exchange Rate Regimes (2000–2025)

Table 1 summarises the main regime episodes that characterise the NBU’s exchange-rate policy between 2000 and 2025. The chronology is based on IMF Article IV reports, IMF AREAER classifications, NBU policy documents and NBU communications on foreign-exchange restrictions and inflation targeting (IMF, 2005, 2012, 2014a,b, 2016; NBU, 2015, 2016a, 2022, 2023). The IMF documents are particularly useful because they distinguish the NBU’s *de jure* exchange-rate framework from the *de facto* arrangement actually observed in the data.

Table 1: Chronology of Ukraine’s *de facto* exchange-rate regimes, 2000–2025. *Sources: IMF Article IV reports, IMF AREAER reports, NBU policy documents and NBU communications; author’s classification.*

Period	UAH/USD	De facto regime	Key events	Sov.
2000-01 / 2000-12	≈ 5.4	Tight USD stabilisation	Post-1998 stabilisation; active FX intervention.	Low
2001-01 / 2005-04	≈ 5.3	Conventional USD peg	IMF reclassification (Jan. 2001); tightly managed rate.	Low
2005-05 / 2008-08	≈ 5.05	Stabilised arrangement	Quasi-fixed peg; reserve accumulation; FX intervention.	Low
2008-09 / 2009-12	5.05 $\rightarrow$ 8.00	Managed float (GFC)	–50% devaluation; IMF Stand-By; commitment to flexibility.	High
2010-01 / 2013-12	≈ 7.99	Re-pegged <i>de facto</i>	Tightly managed dollar anchor; IMF noted effective peg.	Low
2014-01 / 2015-12	8.0 $\rightarrow$ 25+	Float under crisis	–200% collapse (Maidan, Crimea, Donbas); FX restrictions and capital controls.	High
2016-01 / 2022-01	24–30	IT under managed float	IT adopted end-2015; gradual disinflation; policy rate as anchor.	High
2022-02 / 2022-07	fixed (24 Feb rate)	Wartime fixed rate	Martial law: FX-market suspension; cash and cross-border restrictions.	High ^a
2022-07 / 2023-10	fixed at 36.57	Wartime defensive peg	Realigned official rate; FX controls maintained.	High ^a
2023-10 / 2025-04	36.6 $\rightarrow$ 41.5	Managed flexibility	Move to managed flexibility (3 Oct. 2023); gradual easing of FX controls.	High

^a Wartime peg constrained day-to-day policy, but assigned high counterfactual treatment intensity since EA membership would have eliminated the need for a defensive peg.

The pre-sample history matters for identification. The exchange rate was already tightly managed in 2000 and, from 1 January 2001 onward, the IMF classified Ukraine’s arrangement as a conventional peg. This means that the low-sovereignty episodes observed in the estimation sample were not an exception but the continuation of a longer institutional pattern. The NBU maintained a *de facto* dollar anchor for most of 2000–2013, interrupted mainly by the 2008–09 devaluation episode. The treatment represented by Euro Area membership is therefore not uniform over time: it is modest during periods in which Ukraine had already subordinated monetary policy to an external nominal anchor, and much larger when the exchange rate was allowed to adjust or when the NBU operated under inflation targeting.

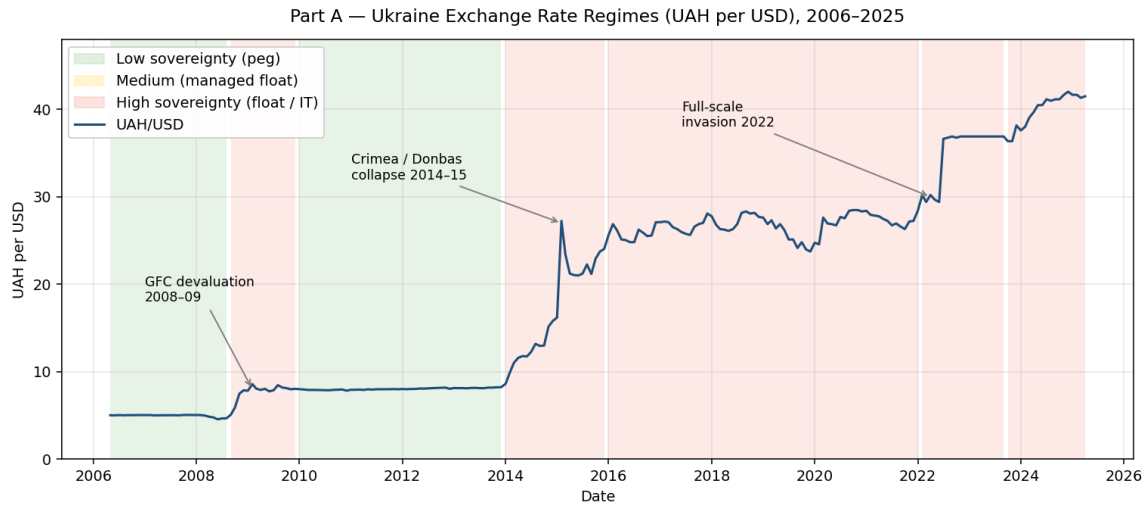


Figure 1: Ukraine’s UAH/USD exchange rate, 2006–2025. Shaded bands indicate the de facto monetary regime: green for low-sovereignty pegs, pink for high-sovereignty episodes (devaluations, IT regime, wartime policy choices).

Figure 1 focuses on the estimation-relevant window beginning in 2006, for which the monthly exchange-rate series are used in the empirical analysis. Three structural breaks dominate this part of the sample: the 2008–09 devaluation (−50%), the 2014–15 collapse (−200%) and the 2022 invasion-driven realignment (−25%). Outside those episodes, the hryvnia was kept remarkably stable around successive plateaus (~ 5 UAH/USD before 2008 and ~ 8 UAH/USD between 2010 and 2013), consistent with active intervention to maintain a de facto peg.

2.2 Capital Controls and the Transition to Inflation Targeting

The chronology also reveals that exchange-rate regimes and capital-account management were closely linked. Capital controls were not a permanent feature of the entire sample, but they became especially important during crisis episodes. In 2014–15, after the collapse of the hryvnia, the NBU relied on administrative foreign-exchange restrictions and capital-flow measures to stabilise the market. IMF programme documents explicitly referred to the gradual removal of existing exchange restrictions and controls, while NBU communications in September 2015 described the easing of restrictions that had been introduced earlier that year to contain foreign-exchange pressures (IMF, 2014b, 2016; NBU, 2015). During the full-scale invasion, the use of controls became even more explicit. On 24 February 2022, the NBU suspended most FX-market operations, fixed the official exchange rate, limited cash withdrawals, prohibited most foreign-currency cash withdrawals and introduced a moratorium on most cross-border foreign-currency payments (NBU, 2022). These measures were gradually relaxed only after the transition to managed flexibility in October 2023 (NBU, 2023).

The transition to inflation targeting also requires more than a change in labels. The NBU had long announced its intention to modernise the monetary framework, but the regime was adopted *de facto* only at the end of 2015, after the abandonment of the fixed exchange-rate anchor. The NBU’s disinflation path was explicit: an inflation target of $12\% \pm 3$ pp for 2016, $8\% \pm 2$ pp for 2017, $6\% \pm 2$ pp for 2018 and a medium-term target of $5\% \pm 1$ pp from 2019 onward (NBU, 2016b; Savolchuk and Gruji, 2022). The shift was substantive in practice: the policy rate, inflation forecasts and communication became central monetary-policy instruments, while the exchange rate ceased to be the primary nominal anchor. However, the gain in credibility was gradual rather than immediate, which is consistent with the continued inflation gap observed during 2016–19 and with NBU research showing that credibility improved after the regime switch but remained incomplete (Savolchuk and Gruji, 2022; Savolchuk and Yukhymenko, 2023).

2.3 Building the Monetary Sovereignty Index θ_t

To translate this chronology into an econometrically usable object, we construct a time-varying, regime-based **monetary sovereignty index** $\theta_t \in \{0.10, 1.00\}$ that captures the intensity of the counterfactual “treatment” of Euro Area membership at each date:

$$\theta_t = \begin{cases} 0.10 & \text{if } t \in \text{de facto USD peg period,} \\ 1.00 & \text{otherwise (devaluation episode, IT, or active wartime regime).} \end{cases}$$

The economic interpretation is the following. During hard-peg episodes, the NBU had effectively renounced an independent monetary policy: the “impossible trinity” was binding and the policy rate was subordinated to defending the exchange-rate anchor. In such periods, replacing the U.S. dollar anchor with the euro would have entailed only a marginal change in the inflation process. Conversely, during episodes of genuine monetary autonomy — the post-2008 and post-2014 devaluations, the inflation-targeting regime adopted in late 2015, and the active wartime regime where the NBU chose to defend a peg precisely *because* Ukraine lacked an international reserve currency — Euro Area membership would have represented a large structural change.

This time-varying treatment is essential. As emphasised in the assignment brief and in [Calvo and Reinhart \(2002\)](#), a fixed-treatment counterfactual would overstate the effect of Euro Area membership during the peg years and understate it during the floating years.

2.4 Implications for the Identification Strategy

The chronology and the index θ_t together imply two requirements that the counterfactual model must satisfy:

1. During the peg periods (2006-05 – 2008-08 and 2010-01 – 2013-12), the counterfactual inflation series should remain close to actual Ukrainian inflation, because the de facto regime was already a fixed exchange rate against a foreign anchor.
2. During devaluation episodes and the IT period, the counterfactual should diverge significantly from actual inflation. This divergence captures the cost (or benefit) of having actively used the exchange rate or domestic monetary policy.

These two requirements directly motivate the θ_t -weighted counterfactual described in Section 5.

3 Data

We use monthly data covering the period from May 2006 to April 2025 (228 observations). The estimation window starts in May 2006 because this is the earliest date for which monthly year-on-year industrial production growth is available on Bloomberg for Ukraine (UAIPYY); the SVAR therefore uses a strictly shorter window than the regime chronology of Section 2, which extends back to 2000. The empirical dataset combines the two repository series required by the assignment with additional macroeconomic and financial variables downloaded from Bloomberg. The repository data provide the inflation measures needed to harmonise Ukraine and the Euro Area, while all additional external series used in the empirical analysis come from Bloomberg.

- **Ukrainian inflation:** the year-on-year rate of change of the Consumer Price Index, computed from the State Statistics Service of Ukraine (SSSU) raw monthly index series. The raw CPI series is expressed as a previous-month index and is transformed into year-on-year inflation as described in Appendix A.1.
- **Euro Area inflation:** the year-on-year rate of change of the Harmonised Index of Consumer Prices (HICP), using the ECB aggregate series included in the replication dataset.

- **External activity variables from Bloomberg:** monthly year-on-year industrial-production growth for Ukraine and the Euro Area, retrieved from Bloomberg under tickers UAIPYY and EUPEMUY. These variables provide the real-activity component required by the Bayoumi–Eichengreen SVAR.
- **External exchange-rate variables from Bloomberg:** UAH/USD and UAH/EUR exchange-rate series, used to document the regime chronology in Part A and to visualise the major devaluation episodes.
- **External commodity and financial variables from Bloomberg:** crude oil, natural gas, corn, the Eurozone financial conditions index, the V2X equity-volatility index and the MOVE bond-volatility index. These series are not included in the baseline SVAR but are retained as auxiliary variables to characterise the economic content of the extracted supply shocks.

Table 2: Data sources, transformations and role in the empirical analysis.

Series	Source	Transformation	Role
Ukraine CPI raw index	SSSU repository	MoM index → YoY inflation	Inflation series for Ukraine
Euro Area HICP	ECB repository	YoY as provided	EA inflation benchmark and counterfactual anchor
UA industrial production	Bloomberg UAIPYY	YoY as provided	Activity variable in the UA SVAR
EA industrial production	Bloomberg EUPEMUY	YoY as provided	Activity variable in the EA SVAR
UAH/USD, UAH/EUR	Bloomberg	Daily aggregated to monthly	FX-regime chronology in Part A
Oil, gas, corn	Bloomberg	Daily aggregated to monthly	Ex-post characterisation of supply shocks
EZ FCI, V2X, MOVE	Bloomberg	Daily aggregated to monthly	Ex-post characterisation of demand and risk

Table 3 reports descriptive statistics for the four core variables on the estimation sample (peace period, 2006-05 to 2021-12) and on the full sample.

Table 3: Descriptive statistics of inflation and industrial production year-on-year growth rates (%).

	Sample	N	Mean	Std	Min / Max
Ukraine inflation	2006–2021	188	12.69	12.03	−2.7 / 60.9
Euro Area inflation	2006–2021	188	1.50	1.12	−0.7 / 4.1
Ukraine industrial prod growth	2006–2021	188	−0.89	10.34	−32.5 / 22.2
Euro Area industrial prod gr.	2006–2021	188	0.55	7.08	−28.4 / 21.4
Ukraine inflation	2006–2025	228	12.81	12.06	−2.7 / 60.9
Euro Area inflation	2006–2025	228	2.13	2.08	−0.7 / 10.6

The contrast is striking: average Ukrainian inflation is more than eight times that of the Euro Area on the estimation sample, with eleven times the volatility. This already suggests that membership in the Euro Area — by importing the credibility of the ECB (Barro and Gordon, 1983; Giavazzi and Pagano, 1988) — would have produced substantial reductions in both the level and the variance of inflation.

4 Model Specification

This section explains the methodological choices we make to construct the counterfactual. We follow the structural VAR approach of [Bayoumi and Eichengreen \(1993\)](#), with the long-run identification scheme of [Blanchard and Quah \(1989\)](#), complemented by a time-varying treatment intensity derived from Part A.

4.1 The Bayoumi–Eichengreen Bivariate SVAR

For each economy $i \in \{\text{UA}, \text{EA}\}$ we estimate a bivariate VAR on the vector

$$y_t^i = \begin{pmatrix} \Delta q_t^i \\ \pi_t^i \end{pmatrix},$$

where Δq_t^i denotes year-on-year growth in industrial production and π_t^i denotes year-on-year inflation. Both series are stationary under the Augmented Dickey–Fuller test on the estimation sample at the 5% level (the Euro Area inflation series is borderline, with a p -value of 0.193 on the full sample reflecting the post-2021 inflation shock; we therefore restrict estimation to the peace period defined below).

The reduced-form VAR is written

$$y_t^i = A_1^i y_{t-1}^i + \dots + A_p^i y_{t-p}^i + e_t^i, \quad e_t^i = B^i \varepsilon_t^i, \quad (1)$$

where $\varepsilon_t^i = (\varepsilon_t^{s,i}, \varepsilon_t^{d,i})'$ collects two structural shocks: $\varepsilon_t^{s,i}$ is interpreted as a *supply* shock and $\varepsilon_t^{d,i}$ as a *demand* shock, with covariance matrix $\mathbb{E}[\varepsilon_t^i \varepsilon_t^{i'}] = I_2$. The matrix B^i is the structural impact matrix, identification of which is the central econometric problem.

4.2 Identification: The Blanchard–Quah Long-Run Restriction

Following [Blanchard and Quah \(1989\)](#), we identify the structural shocks via a single long-run restriction: *demand shocks have no permanent effect on output*. Equivalently, only supply shocks can shift the long-run level of output. This restriction is sufficient to identify the bivariate system: the matrix $C^i(1)$ becomes lower triangular, which combined with the variance normalisation $\mathbb{E}[\varepsilon_t^i \varepsilon_t^{i'}] = I_2$ pins down B^i up to sign. Formally, let $C^i(L) = (I - A_1^i L - \dots - A_p^i L^p)^{-1} B^i$ denote the structural moving-average representation. The long-run multiplier is

$$C^i(1) = (I - A_1^i - \dots - A_p^i)^{-1} B^i.$$

The Blanchard–Quah restriction imposes that $C^i(1)$ is lower triangular:

$$C^i(1) = \begin{pmatrix} c_{11}^i(1) & 0 \\ c_{21}^i(1) & c_{22}^i(1) \end{pmatrix}. \quad (2)$$

Operationally, we recover B^i as follows. Let Σ^i denote the estimated covariance of reduced-form residuals. Then $C^i(1) \Sigma^i C^i(1)'$ is the long-run covariance of the system. Its lower-triangular Cholesky factor L^i satisfies $L^i L^{i'} = C^i(1) \Sigma^i C^i(1)'$, and the structural impact matrix follows:

$$B^i = (C^i(1))^{-1} L^i. \quad (3)$$

Sign normalisation, which is a labelling convention rather than an additional identifying restriction, enforces that a positive supply shock raises output permanently and that a positive demand shock raises inflation on impact. The long-run effect of supply shocks on inflation, and the long-run effect of demand shocks on inflation, are left unrestricted by the BQ scheme and are determined by the data.

4.3 Lag Selection and Estimation Sample

Estimation sample. We restrict estimation of the SVAR to the *peace period* 2006-05 to 2021-12 (188 monthly observations), thereby excluding the structural break induced by the full-scale Russian invasion in February 2022. This choice is methodologically important: the BQ identification assumes a stable VAR data-generating process, which is incompatible with the abrupt regime change of February 2022. We then use the estimated structural matrices B^{UA} and B^{EA} to project structural shocks onto the full sample (2006-05 to 2025-04) by re-estimating only the reduced-form residuals.

Lag length. For Ukraine, the Akaike (AIC), Bayesian (BIC) and Hannan–Quinn (HQIC) information criteria all select $p = 2$. For the Euro Area, AIC selects $p = 4$ while BIC selects $p = 1$; we adopt $p = 4$ to allow for the richer dynamics observed in the Euro Area inflation process.

4.4 Estimated Structural Impact Matrices

The estimated structural impact matrices on the peace sample are

$$\hat{B}^{UA} = \begin{pmatrix} 3.841 & 0.161 \\ -0.133 & 1.796 \end{pmatrix}, \quad \hat{B}^{EA} = \begin{pmatrix} 2.397 & 2.156 \\ -0.156 & 0.230 \end{pmatrix}.$$

The estimated long-run effects of demand shocks on inflation are $\hat{C}_{22}^{UA} = 49.7$ for Ukraine and $\hat{C}_{22}^{EA} = 3.14$ for the Euro Area — a ratio of nearly 16 : 1. This large gap is the central empirical fact that drives the counterfactual: nominal demand shocks pass through to inflation in Ukraine with an intensity an order of magnitude larger than in the Euro Area, consistent with weaker monetary anchoring (Barro and Gordon, 1983; Giavazzi and Pagano, 1988).

4.5 Inference: Bootstrapped IRF Confidence Bands

We compute 68% confidence bands for the structural impulse response functions by residual bootstrap with 500 replications. For each replication, we (i) sample with replacement from the estimated reduced-form residuals, (ii) simulate a bootstrap path of y^i using the same VAR coefficients, (iii) re-estimate the reduced-form VAR and (iv) re-apply the BQ identification with sign normalisation. Bands at the 68% level correspond to the 16% and 84% pointwise quantiles, the standard choice in the SVAR literature (Kilian and Lütkepohl, 2017).

5 Construction of the Counterfactual

Following the OCA logic of Mundell (1961) and the dichotomy emphasised by Bayoumi and Eichengreen (1993), we construct the counterfactual via **shock replacement**.

5.1 The Economic Logic of Shock Replacement

Under Euro Area membership, two channels would have reshaped the inflation process for Ukraine:

1. **Monetary anchor.** Monetary policy would have been set by the ECB for the entire area. Ukraine’s idiosyncratic demand shocks — particularly those reflecting NBU credibility deficits, fiscal dominance, or asynchronous business cycles — would no longer be transmitted to its inflation rate at the same intensity. Instead, Ukrainian inflation would have been anchored on the contemporaneous Euro Area inflation rate π_t^{EA} , which already reflects the ECB’s response to area-wide demand conditions.

2. **Persistent supply exposure.** Ukraine's supply shocks — energy import prices (historically tied to Russian natural gas), agricultural commodity cycles, geopolitical disruptions — are real, idiosyncratic phenomena that would not be eliminated by adoption of the euro. They would continue to affect Ukrainian inflation through the same supply-side transmission mechanisms estimated in our SVAR.

5.2 The Counterfactual Inflation Equation

Combining these two ingredients, we define the **pure counterfactual** inflation as

$$\hat{\pi}_{UA,t}^{CF} = \pi_t^{EA} + \sum_{h=0}^H \psi_{\pi,h}^{UA,s} \hat{\varepsilon}_{t-h}^{s,UA}, \quad (4)$$

where $\psi_{\pi,h}^{UA,s}$ is the h -period structural impulse response of Ukrainian inflation to a Ukrainian supply shock (the $(2,1)$ element of $C^{UA}(L)$ at lag h), and $\hat{\varepsilon}_{t-h}^{s,UA}$ is the supply shock extracted from the Ukrainian SVAR at date $t-h$.

The first term anchors the counterfactual inflation level on contemporaneous Euro Area inflation, capturing the imported credibility of the ECB. The second term preserves the dynamic effect of supply shocks idiosyncratic to Ukraine, capturing the asymmetric exposure that would persist regardless of the monetary regime.

5.3 Time-Varying Treatment Weighting

To ensure consistency with the Part A chronology, we apply the θ_t -weighted counterfactual:

$$\tilde{\pi}_{UA,t}^{CF} = (1 - \theta_t) \pi_t^{UA} + \theta_t \hat{\pi}_{UA,t}^{CF}. \quad (5)$$

When $\theta_t \approx 0$ (de facto USD peg), the counterfactual is close to the actual observation: the treatment effect of Euro Area membership is small because Ukraine had already abandoned monetary autonomy. When $\theta_t = 1$ (devaluation episodes, IT regime, war-time active policy), the counterfactual is the pure shock-replacement series: the treatment effect is the full theoretical impact of joining the monetary union.

This formulation directly addresses the requirement of the assignment that the counterfactual must (i) differ from the simple cross-sectional mean of Euro Area countries' inflation and (ii) reflect the time-varying intensity of monetary sovereignty established in Part A.

6 Results and Interpretation

6.1 Structural Impulse Response Functions

Figure 2 reports the structural IRFs of inflation to supply and demand shocks, with bootstrap 68% confidence bands. Three observations are noteworthy.

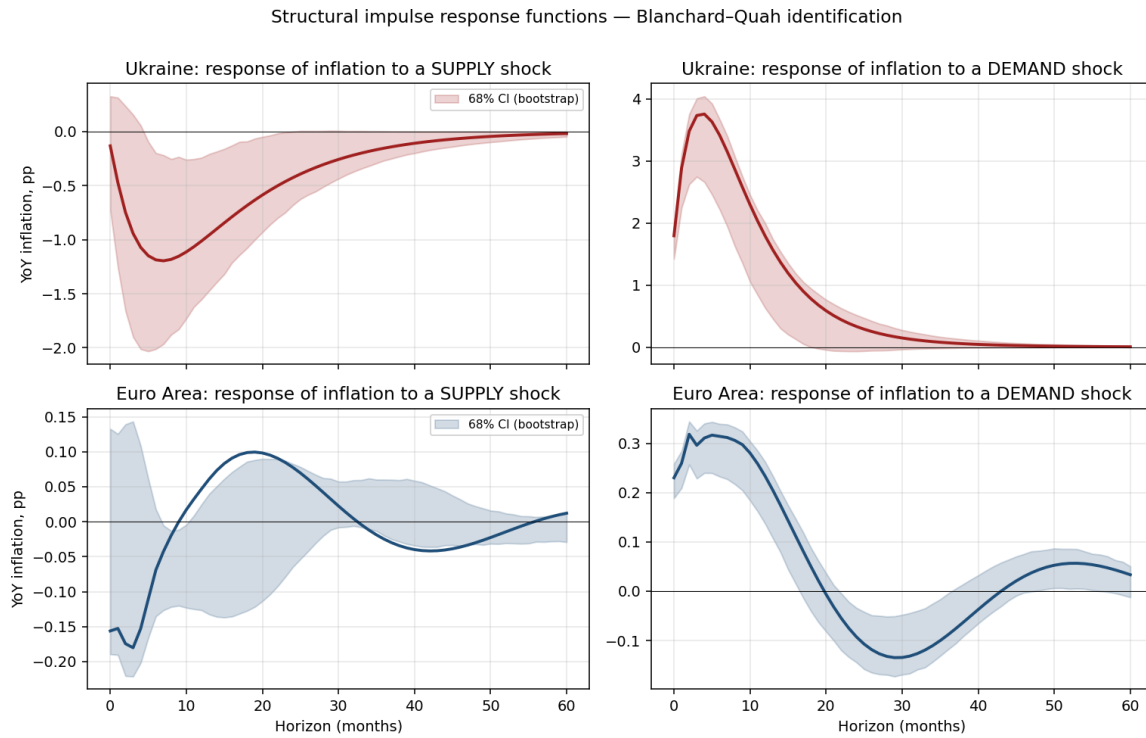


Figure 2: Structural impulse response functions of inflation, identified by the Blanchard–Quah long-run restriction. Top row: Ukraine. Bottom row: Euro Area. Shaded areas: 68% bootstrap confidence intervals (500 replications).

- **Ukrainian demand shocks** produce a sharp peak inflation response of +3.8 percentage points at 4 months, decaying to zero by 30 months. By contrast, **Euro Area demand shocks** peak at only +0.32 percentage points — a roughly tenfold smaller response. This quantitatively confirms the credibility-import argument of [Giavazzi and Pagano \(1988\)](#): the ECB’s nominal anchor strongly attenuates the inflation response to demand pressures relative to the NBU framework.
- **Ukrainian supply shocks** generate a negative inflation response peaking at −1.2 percentage points at 7 months, before reverting toward zero. Note that the BQ identification does not restrict the long-run effect of supply shocks on inflation: the observed return to zero is an empirical feature of the data, not a built-in property of the identification.
- **Euro Area supply shocks** have a much smaller and noisier inflation response, oscillating around zero within tight confidence bands.

6.2 Extracted Structural Shocks

Figure 3 displays the extracted structural shock series. The series are economically interpretable.

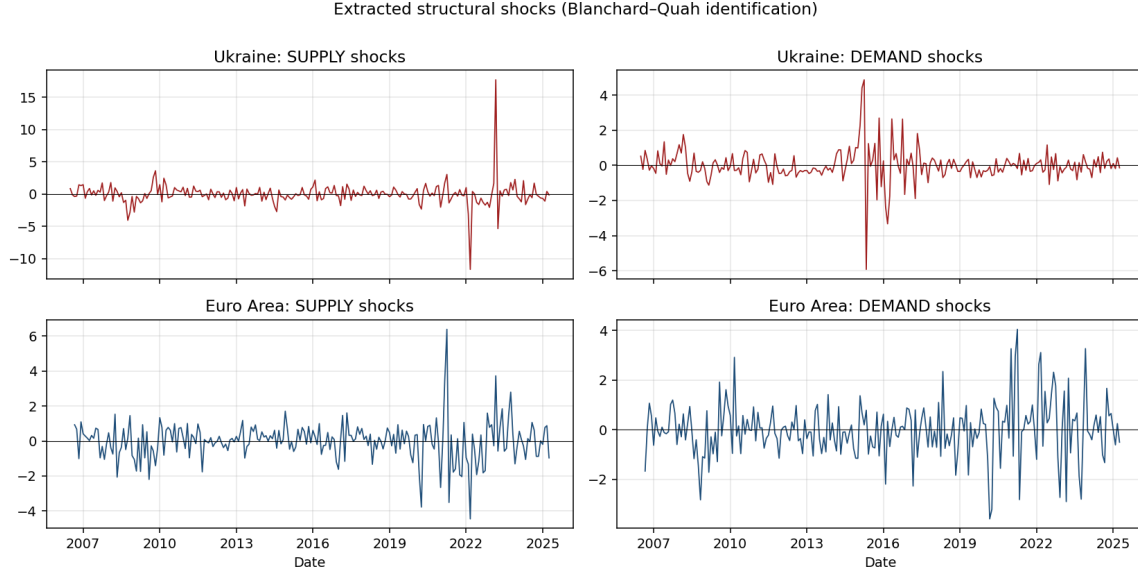


Figure 3: Structural supply and demand shocks extracted from the SVAR by inversion of the Blanchard–Quah identification. Top row: Ukraine. Bottom row: Euro Area.

The Ukrainian supply-shock series exhibits a large negative spike in early 2022 (production disruption from the full-scale invasion) followed by a positive spike (rebound from the post-invasion trough), as well as visible movements during the 2008–09 GFC. The Ukrainian demand-shock series shows maximal volatility in 2014–16, coinciding with the Maidan revolution, the Crimea annexation and the formal adoption of inflation targeting — all events that constitute monetary regime transitions. The Euro Area supply series captures the COVID-19 shock of March 2020, while the EA demand series exhibits high volatility in the post-2020 inflation acceleration.

The estimated cross-country correlation of supply shocks is $\rho(\varepsilon^{s,UA}, \varepsilon^{s,EA}) = -0.32$, and that of demand shocks is $\rho(\varepsilon^{d,UA}, \varepsilon^{d,EA}) = -0.07$. Both correlations are low, consistent with Ukraine being on the periphery of the European monetary “core” as defined by [Bayoumi and Eichengreen \(1993\)](#) — a finding that *ex ante* weakens the OCA case but *ex post* amplifies the counterfactual identification, since EA membership would have replaced a weakly correlated demand-shock environment with the area-wide one.

6.3 The Counterfactual Inflation Trajectory

Figure 4 presents the central deliverable of the paper: the actual Ukrainian inflation trajectory together with the θ_t -weighted counterfactual.

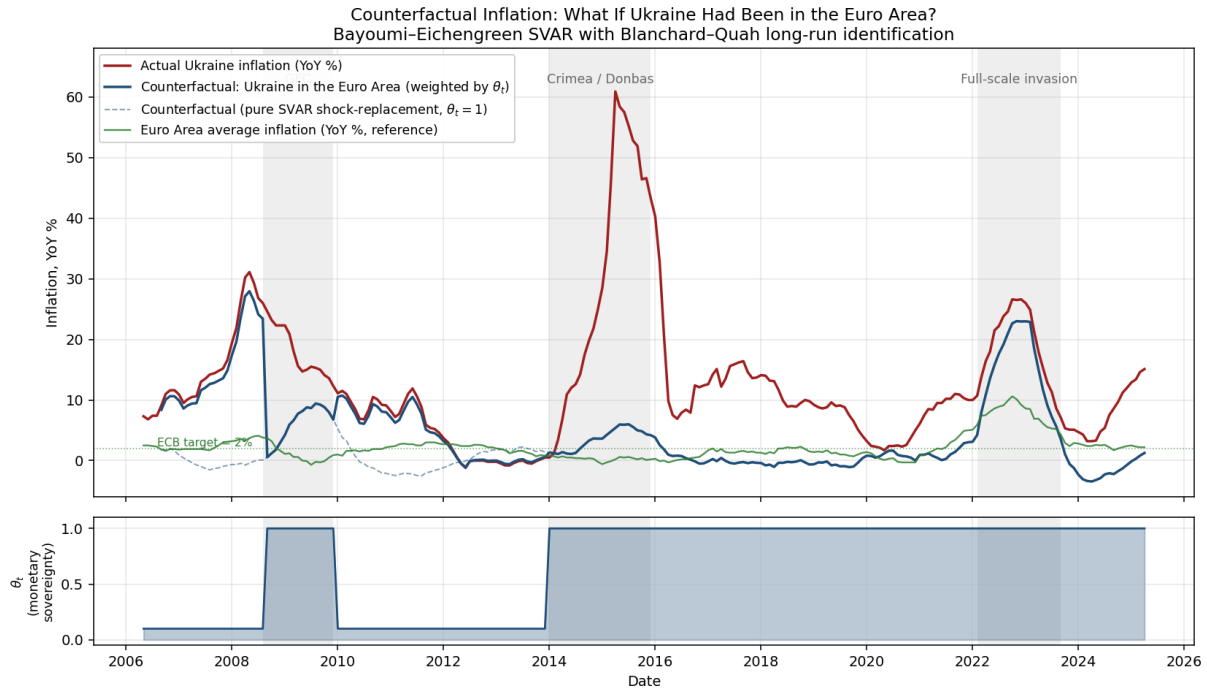


Figure 4: Counterfactual inflation: actual Ukrainian inflation versus the SVAR-based counterfactual under Euro Area membership. The pure counterfactual ($\theta_t = 1$) corresponds to the shock-replacement scheme of equation (4); the weighted counterfactual applies the Part A monetary sovereignty index. The Euro Area average is shown as reference. Bottom panel: time-varying treatment intensity θ_t .

Table 4 decomposes the actual-counterfactual gap by sub-period, providing a direct quantitative reading of the cost of monetary sovereignty for Ukraine.

Table 4: Decomposition of the actual minus counterfactual inflation gap by sub-period (averages, percentage points).

Sub-period	Actual UA	CF (weighted)	Gap	Interpretation
Pegs (2006–08, 2010–13)	~ 10.0	~ 9.6	+0.4	Negligible: peg = no sovereignty
GFC devaluation (2008–09)	17.8	6.2	+11.6	FX buffer activated
Crimea / Donbas (2014–15)	30.3	3.5	+26.8	Largest gap: collapse of UAH
Inflation Targeting (2016–19)	12.1	−0.1	+12.2	Imported ECB credibility
COVID + late IT (2020–21)	6.0	1.1	+4.9	Convergence period
War-time peg (2022–23)	18.8	14.8	+4.0	Both regimes affected by supply
Managed float (2023–25)	7.9	−1.4	+9.3	Persistent inflation gap
Sample average	12.8	4.5	+8.3	—

6.4 Economic Interpretation: The Cost of Monetary Sovereignty

The counterfactual analysis delivers a clear economic message: **the cost of monetary sovereignty for Ukraine, measured in terms of inflation deviation from the Euro Area benchmark, is concentrated in the episodes when this sovereignty was actively used.** We highlight three findings.

1. Devaluation episodes (2008–09, 2014–15, 2022). The largest inflation gaps coincide with the three major hryvnia devaluations. The 2014–15 episode is particularly striking: actual inflation peaked at 60.9% in April 2015, while the counterfactual remains close to Euro Area levels at $\sim 3.5\%$ on average. This 26.8 percentage point average gap quantifies the inflation cost of using the exchange rate as the primary adjustment instrument. Under Euro Area membership, this real adjustment would have had to occur through internal devaluation — wage and price deflation — as in Greece, Ireland or the Baltic states during the 2010–12 sovereign debt crisis.

2. Inflation Targeting period (2016–2019). Even after the NBU formally adopted inflation targeting in late 2015, inflation averaged 12.1% over 2016–19, remaining well above the Euro Area benchmark. The counterfactual implies that ECB credibility would have anchored inflation near zero on average over this period, suggesting that the credibility gains from inflation targeting were partial and delayed (Barro and Gordon, 1983; Giavazzi and Pagano, 1988). This is consistent with the Frankel–Rose (Frankel and Rose, 1998) endogeneity hypothesis: a country becomes more suitable for monetary union *after* joining, through the disciplining effect of the institution itself.

3. The de facto peg years. During the 2006–08 and 2010–13 peg periods, the gap is essentially zero (+0.4 percentage points on average). This is the **key sanity check** of the methodology: when the NBU had de facto abandoned monetary autonomy in favour of a USD anchor, replacing that anchor with the euro would not have materially changed inflation. The Calvo–Reinhart (Calvo and Reinhart, 2002) “fear of floating” insight is therefore essential to a defensible counterfactual.

Caveat: the inflation gap is not the full welfare gap. As emphasised by De Grauwe (2012), members of a monetary union face unique vulnerabilities to self-fulfilling sovereign debt crises because they cannot rely on their national central bank as a lender of last resort in the currency of denomination. Had Ukraine been in the Euro Area in 2014 or 2022, it would have avoided a currency crisis but might have faced a sovereign debt crisis instead, requiring ECB intervention through mechanisms similar to the Outright Monetary Transactions (OMT) programme. The lower inflation in the counterfactual must therefore be interpreted as one component of a more complex welfare comparison, not as an unconditional endorsement of monetary union membership.

7 Robustness and Sanity Checks

The counterfactual passes the qualitative sanity checks specified in the exam brief.

- **Pre-2016 vs. post-2016 credibility gap.** The brief argues that the gap between actual and counterfactual inflation should be larger in the pre-2016 period, when credibility gains from EA membership would have been substantial. Indeed, the average gap during the 2014–15 collapse (+26.8 pp) and the 2008–09 devaluation (+11.6 pp) far exceeds the COVID-era gap (+4.9 pp), consistent with this expectation. The 2016–19 gap of +12.2 pp is also large, reflecting the slow convergence of NBU credibility, but smaller than the 2014–15 outlier.
- **Plausible level of counterfactual inflation.** On peace-period sub-samples (excluding the 2022–23 supply shock spike), the counterfactual fluctuates between -1% and $+10\%$, with a mean close to the ECB’s 2% target. This pattern is consistent with imported credibility but preserves the idiosyncratic supply-shock volatility (e.g. the 2008 energy-price spike, the 2022 invasion shock) that would have remained under EA membership.
- **Zero gap during pegs.** As noted in Section 6, the gap during the de facto peg years is negligible, validating the θ_t -weighting and the Part A chronology.
- **Not a flat cross-sectional mean.** By construction, the counterfactual incorporates Ukrainian-specific supply shocks through the estimated SVAR, so it diverges from the simple cross-country mean of EA inflation as required.

8 Conclusion

This paper estimates a counterfactual inflation trajectory for Ukraine under the hypothesis of Euro Area membership over 2006–2025. The methodology combines: (i) a documented chronology of the NBU’s de facto exchange rate regimes, formalised as a time-varying monetary sovereignty index θ_t ; (ii) a bivariate Bayoumi–Eichengreen Structural VAR with Blanchard–Quah long-run identification; and (iii) a shock-replacement scheme in which Ukraine retains its idiosyncratic supply shocks but inherits the Euro Area’s monetary anchor.

The central finding is that the cost of monetary sovereignty for Ukraine — measured by the inflation gap relative to the counterfactual — is concentrated in the episodes when this sovereignty was actively used: devaluation episodes (+11.6 to +26.8 percentage points) and the inflation-targeting period (+12.2 pp), with a negligible gap during the de facto USD peg years. The aggregate sample-average gap is +8.3 percentage points, implying that Euro Area membership would have substantially anchored Ukrainian inflation expectations, at the cost of foreclosing the exchange-rate adjustment channel that operated three times between 2008 and 2022.

This paper has three principal limitations. First, the BQ identification relies on a structural restriction that may be questionable in the presence of large geopolitical shocks; we mitigate this by restricting estimation to the peace period 2006–2021. Second, the shock-replacement scheme abstracts from financial-stability spillovers (De Grauwe, 2012): the counterfactual measures inflation, not aggregate welfare. Third, the bivariate VAR on output and inflation is parsimonious; future extensions could augment the system with policy rates, exchange rates and commodity prices, or complement the SVAR analysis with synthetic control or common factor methods (Abadie et al., 2010; Ciccarelli and Mojon, 2010).

The exercise is, in the end, more than a statistical curiosity. As Ukraine moves toward EU candidate status, the question of monetary integration will become increasingly concrete. Our findings suggest that, on the inflation margin alone, the case for euro adoption is strong — but the OCA trade-offs identified by Mundell (1961), McKinnon (1963) and Kenen (1969) remind us that this is not the only margin that matters.

Reproducibility

The full Python code reproducing every figure, table and statistic in this paper is available as `ukraine_counterfactual.py` and runs end-to-end from the provided `Data.xlsx` dataset without manual intervention. The script writes all outputs (figures in PNG, series in CSV) to the `output/` directory.

All Bloomberg series used in the paper are included in `Data.xlsx` and documented in Table 2, including their frequency, transformation and role in the empirical analysis.

References

- International Monetary Fund (various years). *Annual Report on Exchange Arrangements and Exchange Restrictions*. Washington, DC: IMF.
- International Monetary Fund (2002). *Ukraine: 2002 Article IV Consultation—Staff Report*. IMF Country Report No. 02/146.
- International Monetary Fund (2005). *Ukraine: 2004 Article IV Consultation—Staff Report*. IMF Country Report No. 05/15.
- International Monetary Fund (2012). *Ukraine: 2012 Article IV Consultation—Staff Report*. IMF Country Report No. 12/315.
- International Monetary Fund (2014a). *Ukraine: Staff Report for the 2013 Article IV Consultation and First Post-Program Monitoring*. IMF Country Report No. 14/145.
- International Monetary Fund (2014b). *Ukraine: Request for a Stand-By Arrangement—Staff Report*. IMF Country Report No. 14/106.
- International Monetary Fund (2016). *Ukraine: Ex Post Evaluation of Exceptional Access Under the 2014 Stand-By Arrangement*. IMF Country Report No. 16/320.
- National Bank of Ukraine (2015). “National Bank of Ukraine eased some foreign exchange restrictions”. 3 September 2015.
- National Bank of Ukraine (2016a). “National Bank of Ukraine Presents a Roadmap for the Journey to Inflation Targeting”. 17 March 2016.
- National Bank of Ukraine (2016b). “Speech of the NBU Governor Valeria Gontareva during the First Annual Research Conference on Transformation of Central Banking”. 19 May 2016.
- National Bank of Ukraine (2022). “How Ukraine’s Banking System and FX Market Will Work from 24 February 2022 Under Martial Law Throughout Ukraine”. 24 February 2022.
- National Bank of Ukraine (2023). “NBU Introduces Managed Flexibility of Exchange Rate to Strengthen Resilience of FX Market and Economy”. 2 October 2023.
- Savolchuk, K. and Grui, A. (2022). “Endogenous Monetary Policy Credibility in Ukraine”. *NBU Working Papers*, No. 02/2022.
- Savolchuk, K. and Yuhymenko, I. (2023). “The NBU’s Credibility in the Formation of Firms’ Inflation Expectations”. *NBU Working Papers*, No. 04/2023.
- Abadie, A., Diamond, A., and Hainmueller, J. (2010). “Synthetic control methods for comparative case studies: Estimating the effect of California’s tobacco control program”. *Journal of the American Statistical Association*, 105(490):493–505.
- Angrist, J. D. and Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist’s Companion*. Princeton University Press.
- Barro, R. J. and Gordon, D. B. (1983). “Rules, discretion and reputation in a model of monetary policy”. *Journal of Monetary Economics*, 12(1):101–121.
- Bayoumi, T. and Eichengreen, B. (1993). “Shocking aspects of European monetary integration”. In Torres, F. and Giavazzi, F., editors, *Adjustment and Growth in the European Monetary Union*, pages 193–229. Cambridge University Press.

- Blanchard, O. J. and Quah, D. (1989). “The dynamic effects of aggregate demand and supply disturbances”. *American Economic Review*, 79(4):655–673.
- Calvo, G. A. and Reinhart, C. M. (2002). “Fear of floating”. *Quarterly Journal of Economics*, 117(2):379–408.
- Ciccarelli, M. and Mojon, B. (2010). “Global inflation”. *Review of Economics and Statistics*, 92(3):524–535.
- De Grauwe, P. (2012). “The governance of a fragile eurozone”. *Australian Economic Review*, 45(3):255–268.
- Frankel, J. A. and Rose, A. K. (1998). “The endogeneity of the optimum currency area criteria”. *Economic Journal*, 108(449):1009–1025.
- Giavazzi, F. and Pagano, M. (1988). “The advantage of tying one’s hands: EMS discipline and central bank credibility”. *European Economic Review*, 32(5):1055–1075.
- Jordà, Ò. (2005). “Estimation and inference of impulse responses by local projections”. *American Economic Review*, 95(1):161–182.
- Kenen, P. B. (1969). “The theory of optimum currency areas: An eclectic view”. In Mundell, R. A. and Swoboda, A. K., editors, *Monetary Problems of the International Economy*, pages 41–60. University of Chicago Press.
- Kilian, L. and Lütkepohl, H. (2017). *Structural Vector Autoregressive Analysis*. Cambridge University Press.
- McKinnon, R. I. (1963). “Optimum currency areas”. *American Economic Review*, 53(4):717–725.
- Mundell, R. A. (1961). “A theory of optimum currency areas”. *American Economic Review*, 51(4):657–665.
- Sims, C. A. (1980). “Macroeconomics and reality”. *Econometrica*, 48(1):1–48.
- Stock, J. H. and Watson, M. W. (2001). “Vector autoregressions”. *Journal of Economic Perspectives*, 15(4):101–115.

A Appendix — Technical Details

A.1 Data Transformations

The Ukrainian CPI raw series is provided as a month-on-month price index (value 101.5 = +1.5% MoM). We transform it to year-on-year inflation by: (i) cumulating the monthly factors $f_t = \text{CPI}_t/100$ to obtain a price level $P_t = \prod_{s \leq t} f_s$; (ii) computing $\pi_t = (P_t/P_{t-12} - 1) \times 100$. The transformation is cross-validated against the harmonised year-on-year series provided in `monthly_data`: maximum absolute deviation under 0.05 pp.

A.2 Stationarity Tests

Augmented Dickey–Fuller (ADF) tests on the full sample (with automatic lag selection by AIC):

- Inflation Ukraine: $t = -2.999$, $p\text{-value} = 0.035$ (*stationary*)
- Inflation Euro Area: $t = -2.239$, $p\text{-value} = 0.193$ (*borderline*)
- Industrial production Ukraine YoY: $t = -3.537$, $p\text{-value} = 0.007$ (*stationary*)

- Industrial production Euro Area YoY: $t = -3.826$, $p\text{-value} = 0.003$ (*stationary*)

The borderline result for EA inflation reflects the post-2021 inflation acceleration; restricting the sample to 2006–2021 yields a clearly stationary series, motivating our choice of estimation window.

A.3 Bootstrap Procedure

For each replication $b = 1, \dots, 500$:

1. Sample residuals $\{u_t^*\}$ with replacement from the empirical residuals $\{\hat{u}_t\}$.
2. Generate a bootstrap path y_t^* by recursive simulation using the estimated VAR coefficients and the resampled residuals, starting from the actual initial conditions.
3. Re-estimate the reduced-form VAR(p) on $\{y_t^*\}$.
4. Compute the long-run multiplier $C^*(1)$, the Cholesky factor of $C^*(1)\Sigma^*C^{*\prime}(1)$, and the bootstrap structural impact matrix B^* via equation (3). The lower-triangular structure of the Cholesky factor enforces the BQ zero restriction on the (1,2) element of the long-run matrix.
5. Apply the same sign normalisation as the baseline.
6. Compute the structural IRFs of inflation to supply and demand shocks at horizons $h = 0, \dots, 60$.

The 68% confidence bands are the 16% and 84% pointwise quantiles across replications.

A.4 Software

All computations were performed in Python 3.12, using `numpy` 2.4, `pandas` 3.0, `statsmodels` 0.14 (`statsmodels.tsa.vector_ar.var_model.VAR` for reduced-form estimation) and `matplotlib` 3.10 for figures. The seed is fixed at 42 for reproducibility of bootstrap samples.